DocChat AI Architecture & Document Loader - Page 1

DocChat AI is an enterprise-grade document intelligence platform powered by Retrieval-Augmented Generation (RA

The DocumentLoader service is responsible for ingesting files from in-memory byte buffers or the local filesystem. F

The system enforces a strict maximum file upload size of 10 MB. Files exceeding 10 MB or featuring unsupported ex

Text chunking is executed by the TextChunker service. Rather than arbitrarily slicing strings at rigid byte boundaries
This overlap prevents semantic cliff effects where critical context, sentence clauses, or named entities are split acros
Each generated chunk is enriched with essential metadata: doc_id, filename, source page number, global chunk ind

DocChat AI Vector Storage & Persistence - Page 3

Once vector embeddings are generated, chunks and their metadata are committed to a persistent ChromaDB collec

The ChromaVectorStore exposes three core methods: add_chunks() for upserting vectorized content with metadata

This storage architecture guarantees high retrieval accuracy, sub-50ms vector query latencies, and deterministic iso

DocChat AI Retrieval & Grounded Generation - Page 4

The GeminiLLM service interfaces with Google Gemini 1.5 Flash to synthesize answers grounded strictly in retrieve

A relevance score threshold of 0.30 is strictly enforced on retrieved chunks. If no retrieved chunk meets or exceeds

By enforcing strict negative constraints, DocChat AI completely prevents hallucinations and avoids speculative asse

In Phase 4, the /chat endpoint streams responses via Server-Sent Events (SSE) with media type text/event-stream.
Conversation memory maintains a sliding window of the last 6 messages per session_id. Prior to vector retrieval, fol